

ELEVANCESKILLS DATA SCIENCE INTERNSHIP

COMPLETE 6-TASK PROJECT REPORT

Project Title: Data Science and Artificial Intelligence – Six Task Internship Project

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Program: B.Tech Artificial Intelligence and Data Science

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1. INTRODUCTION

This project was completed as part of the ElevanceSkills Data Science Internship. The internship provided practical experience in Artificial Intelligence, Data Science, Machine Learning, Deep Learning, Computer Vision, Audio Processing and interactive application development.

The project consists of six practical tasks involving image, audio and machine learning applications. The tasks were implemented and tested primarily using Python and Google Colab, with trained models maintained separately through Google Drive.

The complete project follows the general AI development workflow: Dataset → Data Preprocessing → Feature Extraction → Model Development → Model Evaluation → Model Saving → Prediction → Interactive Application.

2. OBJECTIVES

- To apply Artificial Intelligence and Data Science concepts to practical problems.
- To work with image and audio datasets.
- To perform data preprocessing and feature extraction.
- To develop machine learning and deep learning models.
- To evaluate trained models.
- To save and reuse trained models for prediction.
- To develop practical prediction workflows.
- To build interactive interfaces using Gradio.
- To gain experience in organizing and documenting an end-to-end AI project.

3. TECHNOLOGIES USED

- Python
- TensorFlow and Keras
- Scikit-learn
- OpenCV and MediaPipe
- NumPy and Pandas
- Matplotlib and Seaborn
- Gradio
- Google Colab, Google Drive and GitHub

4. TASK 1 – AGE AND GENDER IDENTIFICATION

Task 1 focused on developing an image-based system for identifying age and gender from facial images. The workflow included facial image preprocessing and development of a deep learning based prediction model.

- Loading and preparing the facial image dataset.
- Preprocessing input images.
- Training and evaluating the age and gender prediction model.
- Saving the trained model for later use.
- Creating a prediction workflow for new facial images.
- Trained model: best_model.keras

5. TASK 2 – SENIOR CITIZEN IDENTIFICATION

Task 2 extended the facial analysis workflow to include senior-citizen identification. The task combined face detection, age prediction and gender prediction to determine whether a detected person belongs to the senior-citizen category.

- Face detection using OpenCV.
- Facial image preprocessing.
- Age prediction and gender prediction.
- Senior-citizen identification.
- Model training and evaluation.
- Saving and loading trained CNN models.
- Testing prediction workflows with images and video.
- Creating an interactive Gradio-based interface.

6. TASK 3 – VOICE AGE, GENDER AND EMOTION ANALYSIS

Task 3 focused on analysing a person's voice using audio features. The system contains a voice-gender classification component and a voice age/emotion prediction component.

- Loading and processing voice recordings.
- Extracting and scaling audio features.
- Voice gender classification using an SVM model.
- Age and emotion prediction using a trained model.
- Determining senior-citizen status from predicted age.
- Creating a voice-based prediction interface.
- Saved files: voice_age_emotion_model.keras, voice_gender_svm.pkl and voice_gender_scaler.pkl.

7. TASK 4 – SIGN LANGUAGE RECOGNITION

Task 4 focused on recognising sign language from hand images using image processing, hand/sign feature extraction and machine learning classification.

- Loading and preprocessing sign language images.
- Detecting hand-related information and extracting features.
- Training and evaluating a classification model.
- Saving and loading the trained model for inference.
- Performing sign prediction from an uploaded image.
- Saved model: sign_language_model.pkl

8. TASK 5 – CAR COLOUR CLASSIFICATION

Task 5 focused on classifying the colour of a car from an input image.

- Loading and preprocessing the car-colour dataset.
- Preparing image classes.
- Training and evaluating a CNN-based classifier.
- Saving the trained model and class mapping.
- Performing prediction on new car images.
- Saved files: car_colour_classifier.keras and car_colour_classes.json.

9. TASK 6 – NATIONALITY-BASED PERSON ATTRIBUTE DETECTION

Task 6 was developed as the final integrated task. The system provides a Gradio interface where a user uploads a person's image and selects a nationality category. Based on the selected category, the system produces the required attribute predictions.

- Indian: Age + Dress Colour + Emotion
- United States: Age + Emotion
- African: Dress Colour + Emotion
- Other: Nationality + Emotion
- The interface includes image upload, nationality selection, image preview, conditional prediction logic and result display.

9.1 EMOTION RECOGNITION

The emotion recognition model was trained for seven classes: Angry, Disgust, Fear, Happy, Neutral, Sad and Surprise.

- Validation Accuracy: 50.07%
- Validation Loss: 1.3371
- Saved model: task6_emotion_model.keras

9.2 DRESS COLOUR CLASSIFICATION

The dress-colour classification component was developed using the required apparel dataset.

- Classes: Black Dress, Blue Dress, Red Dress, White Dress, Yellow Dress
- Total images: 3,136
- Training images: 2,510
- Validation images: 626
- Best validation accuracy: 99.20%
- Final validation accuracy: 99.04%
- Validation loss: 0.0323
- Saved model: task6_dress_colour_model.keras

10. MODEL MANAGEMENT

Trained models were stored separately in Google Drive to keep the GitHub repository organized and lightweight.

- best_model.keras

- senior_citizen_cnn_best.keras
- senior_citizen_cnn_final.keras
- voice_age_emotion_model.keras
- voice_gender_svm.pkl
- voice_gender_scaler.pkl
- sign_language_model.pkl
- car_colour_classifier.keras
- car_colour_classes.json
- task6_emotion_model.keras
- task6_dress_colour_model.keras

Datasets and trained model files are maintained separately from the GitHub repository.

11. PROJECT OUTCOME

The internship resulted in six practical AI-based tasks covering face analysis, age prediction, gender classification, senior-citizen identification, voice-based classification, voice age and emotion prediction, sign language recognition, car colour classification, emotion recognition, dress colour classification, nationality-based conditional prediction and interactive Gradio applications.

The project provided practical experience in the complete development cycle of AI applications, from dataset preparation and preprocessing to model development, evaluation, saving, inference and interactive application development.

12. CONCLUSION

The ElevanceSkills Data Science Internship provided valuable hands-on experience in applying Artificial Intelligence and Data Science techniques to practical problems.

Through the six tasks, the project covered multiple modalities including images, audio and structured prediction workflows.

The internship improved practical understanding of data preprocessing, feature extraction, machine learning, deep learning, computer vision, audio analysis, model evaluation, model persistence and interactive AI applications.

The completed project demonstrates the ability to develop and organize multiple AI solutions using Python and modern Data Science tools.

13. PROJECT RESOURCES

- GitHub Repository: Data-Science-Intern-Project
- Google Drive: Contains the datasets and trained model files used for the internship project.
- Project Notebook: Age_Gender_Identification_Final.ipynb

14. AUTHOR

- Daphne Valentena
- B.Tech Artificial Intelligence and Data Science
- ElevanceSkills Data Science Internship – 2026